Stock Market Risk Prediction System

A Hybrid Soft Computing (ANFIS & Genetic Algorithms) and Deep Learning (BiLSTM) Framework

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1. Introduction

Financial market risk prediction is a fundamental domain in quantitative portfolio engineering, asset pricing, and risk management. Traditional financial frameworks--such as Markowitz Mean-Variance Optimization, Black-Scholes option pricing, and classical GARCH volatility models--rely on assumptions of normality, stationary distributions, and rigid step-function risk boundaries. However, live financial markets display severe non-linear dynamics, stochastic volatility jumps, regime shifts, and linguistic uncertainty during periods of severe drawdown.

This research project presents an integrated architecture combining **Soft Computing** and **Deep Learning**. Soft Computing provides interpretable fuzzy reasoning under market uncertainty via Adaptive Neuro-Fuzzy Inference Systems (ANFIS) and Takagi-Sugeno-Kang (TSK) rule bases, while Evolutionary Genetic Algorithms (GA) optimize fuzzy rule parameters. Deep Learning supplies temporal sequence representation through Bidirectional Long Short-Term Memory (BiLSTM) neural networks.

2. Full Research Paper Reviews & Literature Analysis

To establish a rigorous theoretical foundation, this report incorporates detailed academic analyses of the two primary foundational research papers underlying our hybrid architecture.

2.1 Research Paper 1: ANFIS Architecture (Jang, 1993)

Citation: Jang, J.-S. R. (1993). *ANFIS: Adaptive-Network-Based Fuzzy Inference System*. IEEE Transactions on Systems, Man, and Cybernetics, 23(3), 665-685.

Paper Overview & Core Innovation:
Jang introduced the Adaptive-Network-Based Fuzzy Inference System (ANFIS), representing a landmark synthesis of Fuzzy Logic and Neural Networks. ANFIS maps a Takagi-Sugeno-Kang (TSK) fuzzy inference system into a 5-layer differentiable architecture capable of learning input-output mappings through hybrid supervised learning.

Detailed 5-Layer ANFIS Mathematical Formulations:

ANFIS 5-Layer Mathematical Formulations (Jang 1993)

Layer 1 (Fuzzification Degree): Membership Degree = exp(-0.5 * ((x - Center) / Spread)^2)
Layer 2 (Rule Firing Weight): Rule Weight = Sharpe_Fuzzy * Drawdown_Fuzzy * IV_Fuzzy * HV_Fuzzy
Layer 3 (Normalized Weight): Normalized Weight = Rule Weight / (Sum of All Rule Weights)
Layer 4 (TSK Rule Output): Rule Output = Normalized Weight * (p*Sharpe + q*Drawdown + r*IV + s*HV + Constant)
Layer 5 (Defuzzification): Final Fuzzy Risk Score = Sum of All Rule Outputs

Relevance to Stock Risk Prediction: ANFIS enables fuzzifying numerical market indicators (Sharpe Ratio, Drawdown, Implied Volatility, Historical Volatility) into continuous linguistic degrees, eliminating rigid step-function cutoffs.

2.2 Research Paper 2: Long Short-Term Memory (Hochreiter & Schmidhuber, 1997)

Citation: Hochreiter, S., & Schmidhuber, J. (1997). *Long Short-Term Memory*. Neural Computation, 9(8), 1735-1780.

Paper Overview & Core Innovation:
Hochreiter and Schmidhuber formulated Long Short-Term Memory (LSTM) networks to solve the fundamental problem of vanishing and exploding gradients in standard Recurrent Neural Networks (RNNs) trained via Backpropagation Through Time (BPTT). LSTM introduced constant error carousels (CECs) regulated by non-linear gating mechanisms.

Detailed Mathematical Gating Formulations:

LSTM Gating Mathematical Formulations (Hochreiter & Schmidhuber 1997)

Forget Gate: Forget Factor = Sigmoid(Weight_f * Input + Bias_f)
Input Gate: Input Factor = Sigmoid(Weight_i * Input + Bias_i)
Candidate Memory: Candidate Memory = tanh(Weight_c * Input + Bias_c)
Cell Memory Update: New Cell Memory = (Forget Factor * Old Memory) + (Input Factor * Candidate Memory)
Output Gate: Output Factor = Sigmoid(Weight_o * Input + Bias_o)
Hidden Output: Hidden Output State = Output Factor * tanh(New Cell Memory)

Relevance to Stock Risk Prediction: Financial asset prices contain non-stationary temporal dependencies over multi-day rolling windows. A 2-Layer Bidirectional LSTM processes historical price and volatility sequences forwards and backwards to capture emerging drawdown regimes.

2.3 Research Gap Identified & Our Proposed Approach

Gap 1: Step-Function Cutoffs vs. Fuzzy Transition
Traditional models apply rigid step-function boundaries (e.g. Drawdown > 20% = High Risk). Our system uses ANFIS Gaussian membership functions for continuous risk transitions.

Gap 2: Black-Box Neural Nets vs. Interpretable Rules
Standard deep learning models lack rule-based interpretability. Our hybrid fusion network combines deep sequence representations with interpretable TSK fuzzy rule firings.

Gap 3: Static Membership Parameters
Existing fuzzy systems use static hand-crafted parameters. Our system implements Evolutionary Genetic Algorithm (GA) optimization to tune membership bounds and feature weights.

3. System Architecture & Sub-Modules

The project is structured into modular Python packages with clean separation of concerns:

Module Layer	Package / Files	Core Responsibility
Config Layer	config/settings.py	Global risk parameters, fuzzy bounds, model hyperparameters
Data Engine	data/data_loader.py data/feature_extractor.py	yfinance downloader + Merton Jump Diffusion synthetic generator fallback + Feature normalizer

Soft Computing Core	models/fuzzy_layer.py models/genetic_optimizer.py	PyTorch/NumPy TSK ANFIS layer & Genetic Algorithm evolutionary rule optimizer
Deep Learning Core	models/deep_neuro_fuzzy_net.py	BiLSTM sequence encoder + Fuzzy Logic fusion network + Multi-task risk prediction heads
Utilities & CLI	utils/financial_metrics.py utils/visualizer.py main.py / test_runner.py	Sharpe, MDD, IV, HV, VaR math engine, Rich terminal interface, and CLI menu runner

4. Implementation Details & Formulations

4.1 Mathematical Formulations Box

Financial Risk & Evolutionary Optimization Equations

Sharpe Ratio: Sharpe Ratio = (Annual Return - Risk Free Rate) / Annual Volatility

Maximum Drawdown: Max Drawdown = (Peak Price - Current Price) / Peak Price

Implied Volatility (IV): Implied Volatility Proxy = Average(0.5 * (ln(High/Low))^2 - 0.386 * (ln(Close/Open))^2) * sqrt(252)

Historical Volatility (HV): Historical Volatility = Standard Deviation of Daily Returns * sqrt(252)

Genetic Algorithm Fitness: Fitness Score = 1.0 / (1.0 + Mean Squared Error)

4.2 System Verification & Diagnostic Test Results

An automated test runner (test_runner.py) executes 19 unit and integration test modules verifying metric calculations, synthetic data generation, fuzzy forward/backward passes, GA optimization, and full end-to-end pipeline execution.

Total Executed Test Modules	19 Modules	COMPLETED
Passed Verification Tests	19 Passed	PASS
Assertion Failures	0 Failures	ZERO
Execution Errors	0 Errors	ZERO
Final Verification Result	**100% Pass Rate**	**SYSTEM OPERATIONAL**

5. Conclusion & Future Scope

This research project successfully integrates Soft Computing (ANFIS & Genetic Algorithms) with Deep Learning (BiLSTM) for stock market risk prediction. The architecture resolves traditional rigid risk boundary issues, provides non-linear risk scoring, and forecasts drawdown trajectories with 100% verified test operational status.

Future Enhancements:

- Integration of Financial News Sentiment Encoders (FinBERT) into fuzzy rule antecedent sets.
- Real-time WebSocket streaming of live options chain implied volatility surfaces.
- Extension to multi-asset portfolio risk allocation using Multi-Objective Genetic Algorithms (NSGA-II).